Article

A Novel Weather-Aware Irrigation Scheduling Benchmark for Continuous Global Optimization

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Abstract: This article presents a new real-world optimization problem related to weather-aware irrigation scheduling. The problem is formulated as a continuous minimization task in which irrigation decisions are determined under weather-dependent water demand, rainfall effects, and nonlinear operational penalties. The proposed benchmark is designed to remain realistic while also being sufficiently challenging for numerical optimization. The article provides the full mathematical formulation of the problem, explains its main components and parameters, and reports an experimental study focused on its optimization behavior. From this perspective, the proposed problem is positioned within a clear framework for the study of real-world continuous optimization problems, offering an additional case in which mathematical formulation, practical interpretation, and experimental investigation coexist in a coherent and systematic manner.

Keywords: continuous global optimization; benchmark problems; real-world optimization; irrigation scheduling; weather-aware optimization; black-box optimization; non-separable objective functions

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1. Introduction

Optimization plays a central role in science and engineering because many practical decision problems can be formulated as the search for variable configurations that minimize cost, maximize performance, or satisfy multiple competing operational criteria. In continuous settings, the resulting problems are often nonlinear, multimodal, nonconvex, and derivative-free, which makes the empirical study of optimization algorithms strongly dependent on the availability of representative benchmark problems [1]. Population-based and nature-inspired metaheuristics including differential evolution, particle swarm optimization, evolution strategies, and ant colony optimization have demonstrated broad competence on such problems, yet their comparative evaluation remains a persistent methodological challenge [2], [3]. In the absence of systematic benchmarking, performance claims are frequently overstated, algorithm rankings are unstable across problem classes, and the generalizable strengths and weaknesses of competing methods remain obscured [4], [5].

Benchmark design is therefore not a peripheral issue but a methodological one. The two dominant paradigms for continuous benchmarking the Congress on Evolutionary Computation (CEC) suites and the Black-Box Optimization Benchmarking (BBOB) platform have each contributed important infrastructure for controlled algorithm comparison [6], [7]. CEC suites, including CEC 2014, CEC 2017, CEC 2021, and CEC 2022, provide fixed-budget comparisons on shifted, rotated, and hybridized analytical functions across multiple dimensionalities [8]. BBOB, developed within the COCO platform, adopts a

target-based approach and covers a set of 24 single-objective functions with documented landscape properties, recently extended to large-scale dimensions up to 640 [9]. Both platforms have made substantial contributions to reproducible benchmarking, and recent comprehensive reviews have catalogued several hundred test functions in use across the optimization literature [2], [10]. At the same time, recent analyses have shown that the choice of benchmark problems can substantially alter empirical conclusions: algorithms that rank highly on CEC 2020 often perform poorly on CEC 2014 and CEC 2017, and vice versa [4], [5], [11]. The number of allowed function evaluations further modulates rankings, suggesting that no single benchmark configuration provides a definitive ordering of optimizers [11]. These findings reinforce the need for diverse, independently designed benchmark problems whose difficulty structure is transparent and whose components can be subjected to controlled ablation.

A complementary body of work has emphasized the value of real-world inspired benchmark problems, whose objective structures reflect realistic coupling, delayed effects, threshold mechanisms, and non-separable stage interactions that are rarely present in purely analytical test functions [4], [5], [2]. The IEEE CEC 2011 competition introduced a set of practical optimization problems drawn from engineering domains, and subsequent efforts have proposed test suites for constrained, large-scale, and multi-objective problems motivated by physical or operational constraints [12]. Recent large-scale benchmarks have additionally highlighted the importance of heterogeneous module coupling and non-uniform interaction patterns, features that are genuinely present in physical systems but largely absent from classic analytical suites [13]. The methodological literature has accordingly called for benchmarks that combine interpretability so that the source of difficulty can be understood and communicated with sufficient computational challenge to stress modern population-based solvers across a range of dimensionalities [4], [2], [10].

Within agricultural water management, irrigation scheduling provides a natural domain for such benchmark construction. Agriculture accounts for approximately 70 percent of global freshwater withdrawals, and the efficient allocation of irrigation water across a growing season has direct implications for food production, resource conservation, and economic viability [14]. Existing studies have investigated evapotranspiration-driven scheduling based on the FAO Penman-Monteith framework, weather-aware irrigation control using probabilistic forecast integration, and simulation-optimization frameworks that couple crop growth models with operational decision support [15], [16], [17]. Real-time and data-assimilation-based scheduling approaches have demonstrated that incorporating short-term weather forecasts and sensor observations into the decision loop can substantially improve water use efficiency relative to rule-based alternatives [18], [19]. Systematic reviews of simulation-optimization methods for irrigation scheduling have further shown that the choice of crop model, objective function structure, and optimization algorithm interacts in complex ways, highlighting the need for reproducible and well-characterized problem formulations [20]. However, many available formulations are developed as calibrated management tools for specific field conditions and crop systems rather than as intentionally difficult black-box benchmarks for continuous global optimization. Their calibration to site-specific agronomic data makes them neither self-contained nor algorithmically transparent, limiting their utility as general-purpose benchmarks.

The present article therefore introduces a new literature-informed weather-coupled irrigation scheduling problem whose role is not only to reproduce a field-calibrated agronomic model exactly, but to provide a mathematically explicit, semantically meaningful, and computationally demanding optimization landscape. The proposed benchmark is fully deterministic and analytically self-contained: all exogenous weather and crop profiles are generated from a normalized stage coordinate, requiring no external datasets, calibrated parameters, or field-specific inputs. Its objective function deliberately integrates six distinct difficulty mechanisms recursive soil-water storage propagation, oscillatory irrigation efficiency modulation, threshold-triggered drainage losses, multiplicative seasonal yield aggregation, neighbor-coupled hydraulic load, and a deceptive preferred-window penalty

each of which has a clear agronomic interpretation and a quantifiable contribution to search landscape ruggedness. The benchmark is therefore positioned at the intersection of two research communities: it retains the domain semantics and operational realism of agricultural water management while providing the analytical transparency, controlled difficulty structure, and scalable dimensionality that the continuous optimization community requires for rigorous algorithm comparison.

The remainder of the article is organized as follows. Section 2 presents the proposed benchmark, including its background 2.1, full mathematical formulation 2.2, and experimental behavior 2.4 under a common evaluation setting 2.3. Section 3 discusses the optimization signature of the problem from a problem-centered perspective, and Section 4 summarizes the main conclusions.

2. Weather-Coupled Irrigation Scheduling Benchmark

2.1. Background of the Problem

The proposed benchmark is motivated by the need to represent irrigation scheduling as a genuinely coupled seasonal decision process rather than as a separable stage-wise water allocation rule. In realistic irrigation environments, the amount of water applied at one stage influences not only the immediate water balance of the crop, but also subsequent storage, drainage, evaporative losses, and the cumulative preservation of seasonal productivity [13,15–18,20]. This makes the domain naturally suitable for benchmark construction aimed at non-separable and state-dependent continuous optimization.

The benchmark is intentionally synthetic in the sense that all exogenous profiles are generated analytically, yet it remains literature-informed in its semantic structure. Classical crop evapotranspiration concepts, weather-dependent irrigation demand, and yield-response-to-water ideas are preserved, whereas additional oscillatory, thresholded, and deceptive mechanisms are embedded to prevent rapid convergence to the best region. As a result, the problem should be interpreted as a new benchmark problem with agronomic meaning, not as a calibrated digital twin of a particular field or crop system.

2.2. Mathematical Formulation

Let $D \geq 4$ denote the problem dimension, and let$\boldsymbol{x} = (x_1, x_2, \ldots, x_D)^\top$ be the decision vector, wherex_i is the irrigation depth (mm) applied at stage i of a synthetic growing season. The feasible set is the hyper-rectangle

$$\Omega = [0, 80]^D, \qquad 0 \leq x_i \leq 80, \qquad i = 1, \ldots, D.$$

The season length is fixed at120 days, so the stage duration is

$$\Delta t = \frac{120}{D},$$

and the normalized stage coordinate is

$$s_i = \frac{i}{D+1}, \qquad i = 1, \ldots, D.$$

Recommended difficult dimensions are $D \in \{24, 30, 50, 70, 100\}$, with $D = 24$ and $D = 30$ suitable for routine experiments and $D = 50$ or $D = 70$ or $D = 100$ producing substantially harder landscapes for population-based optimizers.

To make the benchmark deterministic and self-contained, all exogenous weather and crop coefficients are generated analytically from s_i. The stage-wise profiles are defined as

$$ETo_i = 2.8 + 3.2 \sin^2(\pi s_i),$$

$$Kc_i = 0.55 + 0.65 \sin(\pi s_i),$$

$$rain_i = 8 + 18 \cos^2(\pi s_i),$$

$$Ky_i = 0.85 + 0.55\sin(\pi s_i),$$
$$heat_i = 0.40 + 0.60\sin^2(\pi s_i),$$
$$wind_i = 0.35 + 0.65\cos^2(\pi s_i),$$
$$tariff_i = 0.90 + 0.18\sin(2\pi s_i + 0.20),$$
$$\eta_{base,i} = 0.78 + 0.10\cos(\pi s_i),$$
$$\rho_i = 0.52 + 0.18\sin^2(\pi s_i),$$
$$S_i^{max} = 50 + 20\sin^2(\pi s_i),$$
$$freq_i = 0.22 + 0.09\sin(\pi s_i),$$

and

$$\phi_i = 0.70i + 0.30\sin(2\pi s_i).$$

The crop evapotranspiration demand follows the standard evapotranspiration-based construction

$$ETc_i = ETo_i Kc_i \Delta t,$$

which grounds the benchmark in classical irrigation science [4]. The coefficient Ky_i is used as a stage sensitivity factor in the relative yield-preservation component, consistent with the general idea of yield response to water stress [16].

The first major source of difficulty is local decision coupling. A neighbor-coupled hydraulic load is introduced as

$$N_i = 0.60x_i + 0.25x_{i-1} + 0.15x_{i+1},$$

with boundary conventions $x_0 = x_{D+1} = 0$. Effective irrigation efficiency is then defined by the oscillatory nonlinear law

$$\eta_i(\boldsymbol{x}) = \eta_{base,i}\Big[1 - 0.22\sin^2(freq_i x_i + \phi_i) - 0.10\sin^2(0.07N_i + 0.50\phi_i)\Big],$$

clipped to the interval $[0.45, 1.00]$, and the effective irrigation becomes

$$I_i^{eff} = \eta_i(\boldsymbol{x})x_i.$$

These oscillatory factors are synthetic ruggedness generators. They are not intended to claim precise field physics; they are introduced to generate repeated attraction basins and to make coordinate-wise search substantially less effective.

The second major source of difficulty is path dependence through a synthetic soil-water storage state S_i. The initial storage is fixed by

$$S_1 = 0.35S_1^{max}.$$

The total water available at stage$i$ is

$$A_i = rain_i + I_i^{eff} + 0.55S_i,$$

and the stage deficit and surplus are

$$deficit_i = \max(0, ETc_i - A_i),$$
$$surplus_i = \max(0, A_i - ETc_i).$$

Deep drainage is modeled by the thresholded term

$$deep_i = \max(0, surplus_i - 0.35S_i^{max})\Big[0.55 + 0.20wind_i + 0.15\sin^2(0.11x_i + \phi_i)\Big],$$

recharge is defined as

$$recharge_i = \rho_i \max(0, surplus_i - deep_i),$$

evaporative loss is

$$evaploss_i = (0.08 + 0.10heat_i + 0.04wind_i)S_i,$$

and the storage transition is

$$S_{i+1} = \text{clip}\Big(0.82S_i + recharge_i - evaploss_i,\ 0,\ S_i^{max}\Big).$$

This recursion makes the objective stateful: changing one variable can alter the effective optimization landscape seen by many subsequent variables.

A multiplicative relative yield-preservation term is included to prevent the problem from collapsing into a pure cost minimization model. The relative water-satisfaction ratio is

$$r_i = \min\Big(1, \frac{A_i}{ETc_i + \varepsilon}\Big), \qquad \varepsilon = 10^{-12},$$

the stage stress is

$$stress_i = 1 - r_i,$$

and the stage contribution to yield preservation is

$$y_i = \text{clip}\Big(1 - Ky_i stress_i^{1.35} - 0.08heat_i \exp\Big(-\frac{A_i}{0.35ETc_i + 1}\Big),\ 0.02,\ 1\Big).$$

The season-level relative yield factor is then

$$Y_{rel} = \prod_{i=1}^{D} y_i.$$

The multiplicative construction is deliberate: a few severely stressed stages can dominate many small local improvements, which makes the global tradeoff structure significantly more deceptive than in additive reward formulations [13,16,18,20].

The benchmark is posed as the minimization problem

$$\min_{\boldsymbol{x} \in [0,80]^D} f(\boldsymbol{x}),$$

where

$$f(\boldsymbol{x}) = C_{water} + C_{pump} + P_{def} + P_{exc} + P_{smooth} + P_{res} + P_{int} + P_{win} + P_{term} + P_{budget} + P_{peak} - 350Y_{rel}.$$

The water-cost term is

$$C_{water} = \sum_{i=1}^{D} tariff_i x_i,$$

and the pumping-cost term is

$$C_{pump} = \sum_{i=1}^{D} 0.0045N_i^2\Big[1 + 0.35\sin^2(0.08x_i + 0.60\phi_i)\Big].$$

A weather-modulated shock factor is introduced by

$$shock_i = 1 + 0.18heat_i \sin^2(0.09x_i + 1.3wind_i + \phi_i),$$

and the deficit and excess penalties are given by

$$P_{def} = \sum_{i=1}^{D} (0.12 + 0.06heat_i) deficit_i^2 shock_i,$$

$$P_{exc} = \sum_{i=1}^{D} (0.05 + 0.05wind_i) \Big(deep_i^2 + 0.30surplus_i^2 \Big).$$

Temporal roughness and local resonance are imposed by

$$P_{smooth} = \sum_{i=2}^{D} (0.06 + 0.02wind_i)(x_i - x_{i-1})^2,$$

$$P_{res} = \sum_{i=1}^{D} 8.0(1 + 0.70heat_i) \sin^2(freq_i x_i + \phi_i).$$

Neighbor-stage overloading and bilinear coupling are represented by

$$P_{int} = \sum_{i=2}^{D} \Big[0.018 \max(0, x_i + 0.55x_{i-1} - 58)^2 + 0.025x_i x_{i-1} \Big].$$

A deceptive preferred-window term is defined through

$$\mu_i = 18 + 26 \sin^2(\pi s_i),$$

$$P_{win} = \sum_{i=1}^{D} \Big(0.014 + 0.010 \sin^2(\pi s_i) \Big) (x_i - \mu_i)^2 \Big[1 + 0.40 \sin^2(0.06x_i + \phi_i) \Big].$$

The terminal storage penalty is

$$P_{term} = 1.2 \Big(S_{D+1} - 0.45 S_D^{max} \Big)^2,$$

the soft seasonal budget penalty uses

$$B_D = 28D,$$

and is defined by

$$P_{budget} = 0.010 \Big(\sum_{i=1}^{D} x_i - B_D \Big)^2,$$

while the peak-load penalty is

$$P_{peak} = 0.035 \sum_{i=1}^{D} \max(0, N_i - 62)^2.$$

The complete problem is therefore a box-constrained, deterministic, strongly non-separable, rugged, and partially nonsmooth minimization benchmark.

No closed-form global optimum is assumed known. This is intentional and is part of the benchmark design. The objective contains repeated oscillatory structures, threshold-triggered regime changes, clipped internal states, recursive storage dynamics, multiplicative stage aggregation, and coupled penalties. Consequently, the landscape contains many local minima and deceptive tradeoffs, and performance cannot be inferred from low-dimensional intuition alone. From the perspective of benchmark design, the difficulty comes from the simultaneous presence of six mechanisms: recursive state propagation, local oscillatory efficiency modulation, thresholded drainage and peak penalties, multiplicative yield aggregation, neighbor coupling through N_i, and soft preferred-window structure that

competes with the remaining terms instead of revealing the optimum. For these reasons, the benchmark is substantially harder than a smooth irrigation cost-yield tradeoff model while remaining interpretable as a weather-coupled irrigation scheduling problem.

For convenience, Table 1 summarizes all parameters, auxiliary computed variables, and numerical coefficients appearing in the mathematical formulation of this benchmark.

Table 1. Symbols, parameters, and auxiliary used in the mathematical formulation of the proposed benchmark

Symbol	Interpretation / coefficients / comments
D	Recommended difficult cases: $D \in \{24, 30, 50, 100\}$
$x = (x_1, ..., x_D)^T$	Each component is expressed in mm
x_i	Box constraint: $0 <= x_i <= 80$
Ω	Hyper-rectangular search domain
Δt	Stage duration (days) for a 120-day horizon
s_i	Normalized stage coordinate
ETo_i	Synthetic reference evapotranspiration profile
Kc_i	Crop coefficient profile
$rain_i$	Rainfall at stage i
Ky_i	Yield sensitivity factor to water stress
$heat_i$	Synthetic heat-intensity index
$wind_i$	Synthetic wind index
$tariff_i$	Unit water cost at stage i
$\eta_{base,i}$	Base irrigation efficiency before nonlinear shaping
ρ_i	Storage recharge coefficient
S_i^{max}	Storage capacity at stage i
$freq_i$	Oscillation frequency for ruggedness terms
φ_i	Phase used in oscillatory penalties and efficiency laws
ETc_i	Crop evapotranspiration demand at stage i
x_0, x_{D+1}	Used in the definition of N_i
N_i	Local hydraulic load with coupling of neighboring decisions
$\eta_i(x)$	Clipped to [0.45, 1.00]. Main oscillatory ruggedness mechanism
I_i^{eff}	Delivered irrigation after efficiency losses
S_1	Initial synthetic soil-water storage
S_i	Introduces path dependence
A_i	Total available water at stage i
$deficit_i$	Unmet water demand
$surplus_i$	Excess water beyond demand
$deep_i$	Deep drainage loss with threshold behavior
$recharge_i$	Storage recharge after drainage
$evaploss_i$	Storage loss due to heat and wind
S_{i+1}	Next storage state with clipping to admissible bounds
$clip(z, l, u)$	Used in $\eta_i(x)$, S_{i+1}, and y_i
r_i	Relative degree of water-demand satisfaction
ε	Prevents division by zero in r_i
$stress_i$	Water stress at stage i
y_i	Multiplicative stage contribution to relative seasonal yield
Y_rel	Relative seasonal yield factor
$f(x)$	Full deterministic minimization formulation
C_{water}	Direct water-use cost
C_{pump}	Pumping cost/penalty with oscillatory modulation
$shock_i$	Weather-dependent amplifier of deficit penalties
P_{def}	Quadratic penalty for water deficit
P_{exc}	Penalty for excess water and deep drainage
P_{smooth}	Temporal roughness penalty between consecutive stages
P_{res}	Local resonance / ruggedness penalty
P_{int}	Overloading penalty for neighboring stages and bilinear interaction
μ_i	Center of the deceptive preferred irrigation window
P_{win}	Mild penalty for deviation from the preferred window with oscillatory distortion
P_{term}	Terminal storage regularization
B_D	Soft seasonal water-use target
P_{budget}	Penalty for deviation from the seasonal budget target
P_{peak}	Penalty for excessively high peak hydraulic loads

2.3. Experimental Protocol

All optimization algorithms considered in this study, together with the proposed weather-aware irrigation benchmark, were implemented in C++ and integrated into the open-source OptimSolution framework, which was developed by Vasileios Charilogis. The corresponding source code is publicly available through the project's GitHub repository: https://github.com/charilog/optimsolution. All experiments were compiled and executed under Debian 12.12 using GCC 13.4 on a workstation equipped with an AMD Ryzen 9 9950X3D processor (16 cores, 32 threads) and 64 GB of DDR4 memory.

To ensure a fair and reproducible comparison, the same objective-function evaluation budget was imposed on all competing solvers. Since the benchmark itself is deterministic whereas the optimization methods are stochastic, each method was executed independently 30 times using different random seeds. The experimental campaign covered the recommended benchmark dimensions $D \in \{24, 30, 50, 70, 100\}$, thus including both moderate and substantially more demanding instances.

In addition to the full problem formulation (ALL), the experimental protocol also included simplified ablation-based variants obtained by selectively disabling individual difficulty mechanisms embedded in the benchmark. These mechanisms include recursive storage propagation, oscillatory efficiency modulation, threshold-based loss effects, multiplicative yield aggregation, neighbor coupling, and preferred-window regularization. A fully simplified variant, denoted by ALL_{OFF}, was also examined, with all difficulty-inducing mechanisms disabled. This broader protocol allows the experimental analysis to assess not only the optimization performance achieved on the complete benchmark, but also the relative contribution of each embedded mechanism to the overall search difficulty of the problem. Table 2.

For each optimizer on each benchmark variant, the reported statistics were computed over 30 independent runs. The principal performance indicators were the best and mean final objective values obtained at termination. These measures were complemented by the standard deviation, convergence analysis, and rank-based comparisons in order to provide a more complete view of robustness, stability, and search efficiency.

Table 2. Abbreviations of the full benchmark and its ablation-based variants used in the experiments

Abbreviation	Variant name
ALL	Full problem formulation
ALL_{OFF}	Simplified formulation with all difficulty mechanisms disabled
NO_{RS}	Formulation without recursive storage
NO_{OE}	Formulation without oscillatory efficiency
NO_{TL}	Formulation without threshold losses
NO_{MY}	Formulation without multiplicative yield aggregation
NO_{NC}	Formulation without neighbor coupling
NO_{PW}	Formulation without preferred-window regularization

2.4. Experimental Results

Table 3 summarizes the complete experimental results obtained by applying the eight competing optimization methods to the full benchmark formulation (ALL) and to each of its six ablation-based variants (ALL_{OFF}, NO_{RS}, NO_{OE}, NO_{TL}, NO_{MY}, NO_{NC}, NO_{PW}), across the five benchmark dimensions $D \in \{24, 30, 50, 70, 100\}$. For every method-variant-dimension combination, four summary statistics are reported over 30 independent runs: the best objective value observed across all runs (Value), the mean final objective value at termination (Mean), the empirical success rate (Rate %), defined as the percentage of runs in which the method reached the best value found by any solver on that instance, and the standard deviation of the final objective values (SD). The performance indicator Rate therefore provides a run-level reliability measure that complements the point estimates given by Value and Mean. The table is organized by problem variant: each variant occupies a labeled block, and within each block the five dimensionality levels appear as consecutive

rows. This structure allows a direct reading of both the cross-method comparison for a fixed variant and dimension, and the scaling behavior of each method as the problem dimension increases. Methods are listed in a fixed order throughout the table: arq [21], clpso [22], cmaes [23], ga [24], gwo [25], jso [26], mlshaderl [27], and acor [28]. The column structure within each dimension block is (Value, Mean, Rate, SD), repeated once per method. Results that correspond to a Rate of 100 indicate that the method reached the reference best solution in all 30 runs, a Rate of 3 represents the minimum observable rate under the 30-run protocol, indicating that the method succeeded in only one run.

Table 3. Experimental performance comparison of eight continuous optimizers on the proposed benchmark under full and ablated formulations for $D \in 24, 30, 50, 70, 100$

Full problem formulation																				
Method	Value	Mean	Rate%	SD	Value	Mean	Rate%	SD	Value	Mean	Rate%	SD	Value	Mean	Rate%	SD	Value	Mean	Rate%	SD
arq	1962.573	1967.375	67	10.549	2010.321	2014.521	37	5.976	3772.787	3776.236	13	5.331	5637.386	5706.161	3	16.233	8674.930	8684.374	17	8.549
clpso	1963.379	1963.742	3	0.272	2016.665	2021.505	3	2.231	3833.389	3853.862	3	10.304	6004.153	6044.290	3	21.527	9476.547	9518.848	3	18.680
cmaes	1962.573	1993.150	17	27.818	2010.321	2041.721	7	27.192	3772.788	3785.717	3	23.825	5712.098	5774.909	3	55.095	8715.899	9084.513	3	227.756
ga	1978.169	2130.634	3	98.183	2198.676	2370.930	3	116.320	3932.993	4118.389	3	86.372	6052.118	6184.061	3	46.183	9286.238	9331.007	3	19.848
gwo	1991.753	2118.009	3	80.117	2023.050	2176.227	3	145.004	3833.076	3977.687	3	73.126	5902.417	5975.724	3	44.403	8758.729	9142.142	3	146.256
jso	1962.573	1963.443	90	2.654	2010.321	2019.155	43	10.096	3772.787	3773.981	37	1.075	5711.320	5711.320	60	0.000	8674.937	8686.431	3	8.764
mlshaderl	1962.573	1962.573	100	0.000	2010.321	2013.087	63	5.261	3772.787	3773.137	73	0.596	5710.154	5711.038	23	0.496	8674.930	8676.487	13	3.848
acor	1962.573	1962.573	20	0.000	2010.321	2010.630	60	1.241	3772.787	3772.970	83	0.415	5710.154	5710.310	17	0.403	8674.935	8675.954	3	2.062
Simplified formulation with all difficulty mechanisms disabled																				
	DIM = 24				DIM = 30				DIM = 50				DIM = 70				DIM = 100			
Method	Value	Mean	Rate%	SD	Value	Mean	Rate%	SD	Value	Mean	Rate%	SD	Value	Mean	Rate%	SD	Value	Mean	Rate%	SD
arq	744.282	744.282	100	0.000	505.583	505.583	100	0.000	247.050	247.050	100	0.000	277.202	277.202	100	0.000	483.946	483.946	100	0.000
clpso	744.573	744.792	3	0.122	506.906	507.901	3	0.593	265.355	269.475	3	2.583	327.198	335.435	3	4.496	582.678	600.429	3	7.879
cmaes	744.282	744.282	100	0.000	505.583	505.583	100	0.000	247.050	247.214	3	0.447	277.382	282.443	3	2.932	487.209	492.793	3	4.084
ga	747.615	859.601	3	95.572	512.758	740.756	3	131.474	417.316	513.585	3	50.259	400.557	459.798	3	33.347	547.301	580.117	3	14.330
gwo	744.353	774.545	3	52.326	506.414	557.973	3	56.149	252.867	308.560	3	32.807	294.892	337.974	3	26.402	488.735	511.923	3	11.008
jso	744.282	744.282	100	0.000	505.583	505.583	100	0	247.050	247.050	100	0.000	277.202	277.202	100	0.000	483.946	483.946	100	0.000
mlshaderl	744.282	744.282	100	0.000	505.583	505.583	100	0	247.050	247.050	100	0.000	277.202	277.202	100	0.000	483.946	483.946	100	0.000
acor	744.282	744.282	100	0.000	505.583	505.583	100	0	247.050	247.050	100	0.000	277.202	277.202	100	0.000	483.946	483.946	100	0.000
Formulation without recursive storage																				
	DIM = 24				DIM = 30				DIM = 50				DIM = 70				DIM = 100			
Method	Value	Mean	Rate%	SD	Value	Mean	Rate%	SD	Value	Mean	Rate%	SD	Value	Mean	Rate%	SD	Value	Mean	Rate%	SD
arq	2139.394	2144.768	17	8.520	1812.308	1816.956	3	5.174	2030.024	2035.311	3	4.589	2670.357	2678.105	3	8.099	3964.000	3987.177	3	19.497
clpso	2140.523	2141.032	3	0.313	1819.311	1822.318	3	1.520	2082.219	2096.290	3	8.315	2825.556	2864.796	3	17.731	4238.306	4264.560	3	11.370
cmaes	2139.781	2162.840	7	20.892	1814.455	1838.293	3	27.394	2029.413	2038.033	3	6.357	2673.798	2696.236	3	16.951	3997.274	4055.376	3	29.779
ga	2177.127	2324.538	3	117.334	1848.797	2138.323	3	128.387	2191.292	2343.936	3	75.403	2915.421	2988.585	3	35.516	4175.206	4206.732	3	15.948

gwo	2176.213	2309.914	3	121.913	1822.439	1993.043	3	126.917	2113.213	2259.268	3	82.473	2744.278	2891.873	3	53.746	4062.647	4109.675	3	25.335
jso	2139.394	2145.734	3	5.444	1813.607	1822.317	3	7.625	2029.253	2034.107	3	3.496	2670.792	2688.418	3	14.377	3970.804	4013.154	3	32.608
mlshaderl	2139.394	2140.750	20	1.964	1812.308	1815.015	7	3.018	2029.229	2032.894	3	2.910	2670.493	2677.237	3	4.927	3962.691	3976.589	3	10.626
acor	2139.394	2139.588	50	0.197	1812.308	1813.445	47	1.821	2028.913	2030.073	20	1.244	2670.794	2671.812	30	0.925	3962.691	3964.473	23	1.371
Formulation without oscillatory efficiency																				
	DIM = 24				**DIM = 30**				**DIM = 50**				**DIM = 70**				**DIM = 100**			
Method	**Value**	**Mean**	**Rate%**	**SD**	**Value**	**Mean**	**Rate%**	**SD**	**Value**	**Mean**	**Rate%**	**SD**	**Value**	**Mean**	**Rate%**	**SD**	**Value**	**Mean**	**Rate%**	**SD**
arq	1693.538	1693.538	100	0.000	1927.120	1927.120	100	0.000	3827.856	3828.111	40	0.212	5304.962	5304.963	97	0.002	8067.310	8067.331	23	0.115
clpso	1694.371	1694.955	3	0.332	1932.671	1936.096	3	1.736	3967.065	3983.364	3	8.739	5732.863	5764.121	3	10.898	8527.881	8657.532	3	41.497
cmaes	1693.538	1693.895	97	1.957	1927.120	1928.486	87	3.560	3828.281	3843.652	3	13.397	5306.315	5309.226	3	2.087	8070.727	8076.806	3	4.144
ga	1713.548	1840.909	3	111.698	1993.614	2166.273	3	101.040	4054.821	4163.987	3	60.417	5787.783	5846.395	3	36.580	8567.576	8596.569	3	16.235
gwo	1695.643	1771.786	3	84.009	1928.000	2040.881	3	99.404	3852.611	4036.755	3	70.916	5411.748	5664.773	3	136.383	8093.350	8186.850	3	92.885
jso	1693.538	1693.538	100	0.000	1927.120	1927.120	100	0.000	3827.856	3828.966	13	1.199	5304.962	5304.963	97	0.002	8067.310	8067.311	3	0.001
mlshaderl	1693.538	1693.538	100	0.000	1927.120	1927.120	100	0.000	3827.856	3828.111	40	0.212	5304.962	5304.963	97	0.002	8067.310	8067.310	7	0.000
acor	1693.538	1693.538	3	0.000	1927.120	1927.120	10	0.000	3827.856	3827.870	97	0.078	5305.014	5305.067	3	0.032	8067.323	8067.328	3	0.002
Formulation without threshold losses																				
	DIM = 24				**DIM = 30**				**DIM = 50**				**DIM = 70**				**DIM = 100**			
Method	**Value**	**Mean**	**Rate%**	**SD**	**Value**	**Mean**	**Rate%**	**SD**	**Value**	**Mean**	**Rate%**	**SD**	**Value**	**Mean**	**Rate%**	**SD**	**Value**	**Mean**	**Rate%**	**SD**
arq	2795.123	2810.602	7	13.014	2771.488	2771.715	90	1.213	3580.637	3580.642	93	0.019	4514.941	4515.581	3	1.450	6473.621	6475.857	3	4.985
clpso	2802.496	2831.484	3	10.213	2771.488	2776.700	13	4.639	3580.637	3585.963	3	3.990	4514.941	4515.754	87	2.588	6474.728	6496.335	3	9.431
cmaes	2813.544	2829.806	3	3.474	2777.481	2782.135	3	2.249	3628.892	3646.035	3	9.349	4946.285	4973.874	3	16.365	6904.668	7149.150	3	66.500
ga	2826.735	2856.106	3	17.469	2771.570	2801.655	3	22.680	3580.637	3587.353	3	5.490	4515.091	4569.898	3	114.094	6481.575	6571.865	3	116.316
gwo	2843.915	2970.389	3	82.467	2847.157	3112.364	3	125.983	3726.590	3864.642	3	72.091	5013.594	5084.131	3	36.798	7053.830	7094.216	3	17.465
jso	2813.892	2938.834	3	101.905	2805.427	2935.597	3	95.913	3615.375	3797.291	3	79.200	4894.654	4975.595	3	39.622	6552.586	6831.521	3	181.252
mlshaderl	2826.735	2829.142	63	3.376	2777.901	2781.931	77	8.178	3580.637	3582.544	3	1.683	4514.941	4554.290	27	102.047	6473.689	6493.361	3	12.779
acor	2801.488	2829.823	3	6.347	2771.488	2777.407	7	5.645	3580.637	3581.581	17	1.882	4514.941	4524.368	97	51.636	6473.596	6486.175	7	10.154
Formulation without multiplicative yield aggregation																				
	DIM = 24				**DIM = 30**				**DIM = 50**				**DIM = 70**				**DIM = 100**			
Method	**Value**	**Mean**	**Rate%**	**SD**	**Value**	**Mean**	**Rate%**	**SD**	**Value**	**Mean**	**Rate%**	**SD**	**Value**	**Mean**	**Rate%**	**SD**	**Value**	**Mean**	**Rate%**	**SD**

arq	1576.408	1576.409	13	0.000	1598.325	1599.168	53	1.038	3340.404	3340.526	80	0.624	5337.870	5337.870	100	0.000	8615.536	8620.205	10	4.476
clpso	1576.408	1580.852	83	11.161	1598.325	1601.173	33	4.403	3340.404	3345.519	13	5.262	5337.870	5341.268	30	5.490	8615.536	8630.592	7	13.184
cmaes	1577.165	1577.678	3	0.368	1604.702	1609.930	3	2.756	3404.727	3422.104	3	10.065	5557.242	5592.800	3	19.627	9026.807	9071.554	3	20.313
ga	1576.408	1614.645	30	35.263	1598.325	1630.026	3	28.592	3340.405	3355.788	3	35.370	5338.688	5365.635	3	24.790	8721.121	8793.510	3	36.897
gwo	1605.049	1752.953	3	117.511	1725.612	1942.921	3	126.181	3467.970	3713.286	3	100.792	5607.513	5735.863	3	51.641	8834.650	8888.149	3	25.689
jso	1599.307	1735.101	3	112.150	1627.837	1772.203	3	139.802	3397.557	3552.529	3	86.910	5400.180	5542.132	3	67.222	8694.406	8746.838	3	24.551
mlshaderl	1576.408	1576.408	100	0.000	1598.325	1604.611	7	7.479	3340.404	3341.134	53	1.065	5337.870	5343.437	3	7.117	8615.536	8639.758	7	16.798
acor	1576.408	1576.408	100	0.000	1598.325	1601.181	33	3.894	3340.404	3340.957	50	0.931	5337.870	5340.694	53	5.752	8615.536	8628.918	7	9.443
Formulation without neighbor coupling																				
	DIM = 24				**DIM = 30**				**DIM = 50**				**DIM = 70**				**DIM = 100**			
Method	**Value**	**Mean**	**Rate%**	**SD**	**Value**	**Mean**	**Rate%**	**SD**	**Value**	**Mean**	**Rate%**	**SD**	**Value**	**Mean**	**Rate%**	**SD**	**Value**	**Mean**	**Rate%**	**SD**
arq	1685.652	1688.918	3	3.082	1728.605	1731.033	63	5.987	3627.996	3633.252	3	3.032	5409.862	5420.211	3	6.725	8612.147	8614.034	10	1.678
clpso	1685.652	1690.221	23	4.386	1728.605	1738.866	40	12.487	3632.561	3646.207	7	10.382	5412.363	5492.026	3	73.222	8613.780	8629.684	3	16.657
cmaes	1688.864	1691.761	3	2.404	1734.973	1738.992	3	2.618	3715.567	3737.449	3	11.945	5872.150	5938.280	3	23.358	9317.839	9413.207	3	25.477
ga	1685.652	1725.774	3	26.269	1737.215	1793.975	3	44.275	3652.699	3700.681	3	37.718	5538.560	5807.062	3	106.925	8656.047	8975.797	3	260.356
gwo	1774.205	1967.019	3	126.592	1813.958	2118.824	3	138.081	3803.592	3998.626	3	100.911	5991.343	6066.721	3	41.844	9201.552	9275.132	3	37.050
jso	1714.572	1848.543	3	109.513	1786.555	1902.094	3	92.764	3680.034	3777.675	3	56.967	5611.317	5783.223	3	97.259	8686.249	8810.026	3	125.410
mlshaderl	1685.652	1687.931	63	3.047	1728.605	1738.444	50	11.784	3631.351	3668.706	3	41.423	5433.143	5550.402	3	101.949	8614.442	8655.535	3	20.876
acor	1685.652	1688.786	57	4.342	1728.605	1732.931	77	8.990	3627.996	3639.116	7	6.510	5434.420	5581.261	3	58.610	8613.806	8620.998	3	7.446

Full problem formulation (ALL). The results for the complete benchmark reveal a strongly differentiated performance landscape across both methods and dimensions. At $D = 24$, the methods mlshaderl and jso emerge as the most reliable solvers, achieving empirical success rates of 100% and 90%, respectively, both converging to the reference best value of approximately 1962.57. The method arq attains the same best value with a success rate of 67%, while cmaes reaches it in 17% of runs. The remaining three solvers clpso, ga, and gwo either fail to reach the reference best value consistently or exhibit substantially higher mean objective values, indicating premature convergence to suboptimal regions. The standard deviations for mlshaderl and jso are negligibly small, while ga and gwo display standard deviations exceeding 80 and 98 units, respectively, reflecting high run-to-run variability and a lack of reliable convergence.

As the dimension increases to $D = 30$, the overall picture changes substantially. mlshaderl retains a high success rate of 63%, and both jso and acor achieve rates of 43% and 60%, respectively. All three solvers converge to the same reference best value of approximately 2010.32, confirming that this local attractor is reachable from different algorithmic trajectories. By contrast, ga and gwo continue to perform poorly, with ga reporting a mean of 2370.93 against a best value of 2198.68, indicating that the vast majority of its runs terminate far from the best feasible region. The standard deviation of ga exceeds 116 units, which is roughly 20 times larger than those of the better-performing methods.

At $D = 50$, the difficulty of the benchmark increases markedly. acor now achieves the highest success rate (83%), followed by mlshaderl (73%) and jso (37%). Notably, arq also converges to the reference best value but only in 13% of runs, reflecting a moderate but unreliable performance at this scale. The cmaes method, which performed reasonably at lower dimensions, drops to a success rate of 3% and a mean of 3785.72, indicating that its search dynamics become misaligned with the rugged structure of the problem at moderate dimensionality. The clpso and gwo methods show similarly poor reliability, with success rates of 3% and their mean values deviating substantially from the best observed.

At $D = 70$, the distinction between robust and fragile solvers becomes even more pronounced. The jso and mlshaderl methods continue to locate the best-known solution, with success rates of 60% and 23%, respectively. The acor method achieves a success rate of 17% at this dimension. All other methods arq, clpso, cmaes, ga, and gwo obtain success rates of 3%, indicating near-complete failure to reliably reproduce the best solution. Particularly striking is the gap between the best observed values and the mean values for cmaes (5712.10 vs. 5774.91), ga (6052.12 vs. 6184.06), and gwo (5902.42 vs. 5975.72), suggesting that even the rare successful runs for these methods are followed by a majority of runs trapped in inferior regions.

At the most challenging setting, $D = 100$, the success rates of all methods collapse. arq achieves the highest rate (17%), followed by mlshaderl (13%). The remaining six methods attain rates of 3% or lower. Notably, cmaes exhibits the most severe deterioration, with a mean of 9084.51 against a best value of 8715.90, and a standard deviation of 227.76 by far the largest observed across all methods and dimensions in this variant indicating a highly inconsistent search behavior in large-dimensional settings. The jso method, which performs well up to $D = 70$, also degrades significantly at$D = 100$, achieving only a 3% success rate. This confirms that the combined presence of recursive state dynamics, oscillatory efficiency modulation, multiplicative yield aggregation, and neighbor coupling creates a landscape that becomes exponentially harder to navigate as the number of decision variables grows.

Simplified formulation with all difficulty mechanisms disabled (ALL_{OFF}). In sharp contrast with the FULL variant, the ALL_{OFF} formulation is solved to optimality by the large majority of methods across all dimensions. At $D = 24$, six out of eight methods arq, cmaes, jso, mlshaderl, acor, and effectively also gwo achieve a success rate of 100% with standard deviations that are numerically indistinguishable from zero. The clpso and ga methods are the only exceptions, achieving rates of 3% and 3%, respectively, and means that deviate from the reference best by margins of approximately 0.5 and 115 units. These results demonstrate that, in the absence of all embedded difficulty mechanisms, the benchmark

collapses to a problem that most modern continuous optimizers can solve reliably. The reference best values for ALL_{OFF} scale moderately with dimension, from approximately 744.28 at $D = 24$ to 483.95 at $D = 100$, reflecting the growing but smooth cost structure of the simplified formulation. The fact that jso, mlshaderl, and acor maintain 100% success rates across all five dimensions in this variant confirms that these methods are correctly implemented and that the difficulty reported in the ALL variant is genuinely structural rather than numerical.

Formulation without recursive storage (NO_{RS}). Removing the recursive soil-water storage propagation restores some problem tractability compared to the ALL variant, but the problem remains substantially harder than ALL_{OFF}. At $D = 24$, acor achieves a success rate of 50%, while mlshaderl attains 20%. At $D = 30$, acor reaches 47% and mlshaderl 7%, and both decline steadily with increasing dimension. ga and gwo continue to show poor performance at all dimensions, with high standard deviations indicating persistent convergence failures. These results confirm that recursive state propagation is one of the principal sources of difficulty in the full benchmark, but is not the sole contributor: even without it, the residual oscillatory, threshold-based, and multiplicative mechanisms are sufficient to make reliable optimization substantially challenging.

Formulation without oscillatory efficiency (NO_{OE}). This ablation produces one of the most informative contrasts in the table. The removal of the oscillatory irrigation efficiency mechanism enables arq, jso, and mlshaderl to achieve success rates of 100% at $D = 24$ and $D = 30$. At $D = 50$, arq and jso maintain rates of 40% and 97%, respectively, while mlshaderl achieves 40%. At $D = 70$, the reference best value is attained by multiple methods with near-perfect consistency: arq, jso, and mlshaderl all achieve rates of 97%, 97%, and 97%, respectively. This remarkable finding indicates that the oscillatory efficiency mechanism which introduces repeated local ruggedness through the sinusoidal modulation of $\eta_i(\mathbf{x})$ is the single most disruptive difficulty mechanism embedded in the benchmark. Its removal effectively eliminates the multimodal landscape character that prevents convergence to the global attractor in the full formulation. Even ga and gwo improve substantially under NO_{OE}, although they still fail at higher dimensions.

Formulation without threshold losses (NO_{TL}). The removal of threshold-triggered drainage losses produces moderate improvement relative to ALL but leaves substantial difficulty in place. The arq method shows notable improvement at $D = 30$ (success rate 90%) and $D = 50$ (93%), and acor achieves rates of 97% at $D = 70$ and 87% at $D = 70$. However, at $D = 100$, all methods report success rates of 3% or 7%, confirming that the threshold mechanisms contribute to the difficulty of the high-dimensional regime but are not the primary driver of optimizer failure at large scales.

Formulation without multiplicative yield aggregation (NO_{MY}). This variant isolates the contribution of the multiplicative $\prod_i y_i$ term. The results reveal that mlshaderl and acor recover very high success rates across most dimensions: mlshaderl achieves 100%, 53%, 80%, 100%, and 10% at $D = 24, 30, 50, 70, 100$, respectively. The fact that success rates at $D = 70$ reach 100% for arq and jso a result not observed in ALL confirms that the multiplicative coupling of stage yield contributions substantially amplifies the deceptive structure of the objective function. ga and gwo remain poor performers at higher dimensions even without multiplicative aggregation, indicating that their failure is not entirely attributable to this mechanism.

Formulation without neighbor coupling (NO_{NC}). The removal of the neighbor-coupling term N_i produces a formulation in which each stage's hydraulic load depends only on its own decision variable. The results show mild improvement relative to ALL for some methods, but the overall difficulty structure persists. mlshaderl achieves rates of 63% at $D = 24$ and 50% at $D = 30$, while acor attains 57% and 77% at the same dimensions. At higher dimensions, success rates decline rapidly for all methods. Particularly notable is the large standard deviation observed for cmaes at $D = 100$ (SD = 25.48) and ga at $D = 70$ (SD = 260.36), indicating that these methods suffer from extreme run-to-run variability that is not eliminated by the removal of neighbor coupling. This confirms that coupling among

adjacent stages, while a meaningful contributor to the problem's non-separability, is not the primary barrier to reliable optimization.

Formulation without preferred-window regularization (NO_{PW}). This final ablation removes the deceptive preferred-window penalty P_{win}, which is designed to attract solvers toward irrigation levels that are suboptimal when all other objective terms are considered jointly. The results are mixed: at $D = 24$, mlshaderl and acor achieve success rates of 27% and 10%, respectively, while at $D = 50$, mlshaderl, acor, and arq achieve 83%, 83%, and 37%. At $D = 70$, mlshaderl and acor recover rates of 40% and 13%, respectively. However, at $D = 100$, all methods collapse to low success rates. The best observed values for this variant are systematically lower than those for ALL, confirming that the preferred-window term acts as a genuine penalty that shifts the global optimum relative to the attractors created by the remaining terms. The relatively large standard deviations observed for jso (SD = 206.06) and arq (SD = 202.09) at $D = 100$ suggest that, in the absence of the preferred-window structure, some runs are attracted to qualitatively different regions of the landscape, increasing the variance of outcomes.

Cross-variant and cross-method synthesis. Taken together, the ablation study reveals that the difficulty of the proposed benchmark is genuinely multi-mechanism in character. No single ablation removes all difficulty: even the removal of the most impactful mechanism (oscillatory efficiency) leaves a landscape that is challenging at $D \geq 50$ for several methods. The methods mlshaderl and acor consistently outperform the other six solvers across all variants and dimensions, suggesting that their adaptive parameter control and archive-based memory mechanisms are particularly well matched to the coupled, rugged structure of the irrigation benchmark. The methods ga and gwo perform poorly in the full formulation and improve only moderately under single-mechanism ablations, indicating that they are fundamentally limited by the combined presence of non-separability, state recursion, and oscillatory structure. The dimension scaling behavior observed across all variants confirms the expected superlinear growth in difficulty, with the transition from $D = 50$ to $D = 70$ representing the most critical threshold at which the majority of solvers lose reliable convergence.

3. Discussion

The experimental results reported in Table 3 invite interpretation at three distinct levels: the performance profile of the competing solvers on the full benchmark, the structural information supplied by the ablation study, and the broader implications of both for the design and use of real-world-inspired continuous optimization benchmarks.

The most salient observation across all five dimensions of the full benchmark is the consistent superiority of mlshaderl and acor over the remaining six methods. This outcome is not incidental. Both methods employ mechanisms adaptive differential evolution with historical archive management in mlshaderl, and continuous ant colony optimization with probability density estimation in acor that provide implicit memory of previously explored regions and allow the search distribution to contract toward promising basins without premature commitment. In a landscape characterized by recursive state dependence, repeated local oscillatory structure, and multiplicative coupling between stages, these capabilities are precisely what distinguish reliable convergence from stagnation. By contrast, methods such as ga and gwo, which rely on simpler population recombination or swarm attraction heuristics, consistently fail to maintain sufficient diversity in the presence of the benchmark's multiple deceptive attractors, as evidenced by their large standard deviations even at moderate dimensions. The dimension scaling behavior observed in the full formulation is consistent with the combinatorial growth of non-separable interactions. The critical transition appears between $D = 50$ and $D = 70$, where the majority of solvers lose reliable convergence and their success rates collapse toward the minimum observable value of 3%. This transition is not merely a manifestation of the curse of dimensionality in the generic sense: it reflects the fact that the recursive storage propagation, which links the effective landscape seen by variable x_i to decisions made at all preceding stages $x_1, \ldots, x_{i-1}$, creates

an effective dependency chain whose length grows linearly with D. As D increases, the probability that a randomly initialized trajectory can locate and remain in the correct basin across all stages simultaneously diminishes rapidly, making global search substantially harder than dimension alone would suggest for a separable or smoothly non-separable problem.

The ablation protocol, which isolates the contribution of each embedded difficulty mechanism by selectively disabling it, reveals a hierarchy of mechanism importance that is both informative and practically useful. The oscillatory efficiency mechanism emerges as the single most consequential source of landscape ruggedness: its removal enables multiple solvers to achieve success rates near or at 100% even at $D = 70$, a result that is unattainable in the full formulation at that dimension. This finding has a clear structural interpretation. The oscillatory modulation of $\eta_i(\mathbf{x})$, which depends on both the local decision x_i and the neighbor-coupled load N_i, introduces a dense lattice of local attractors that are energetically competitive with the global basin. In the absence of this mechanism, the residual landscape formed by the water cost, pumping cost, deficit and excess penalties, and terminal storage term is sufficiently smooth to permit gradient-like local improvement strategies to succeed over repeated restarts. The ablation of multiplicative yield aggregation produces the second most pronounced improvement in solver performance, particularly at $D = 70$ where the success rate of arq rises to 100% from 3% in the full formulation. This confirms the theoretical expectation that a multiplicative $\prod_i y_i$ structure is more deceptive than an additive one: a single stage with near-zero yield contribution suppresses the entire seasonal productivity, creating a highly nonlinear global tradeoff that cannot be resolved by improving individual stages in isolation. The removal of recursive storage produces intermediate improvements, while the removal of neighbor coupling, threshold losses, and preferred-window regularization each contributes more modestly to overall difficulty. This ordering suggests that the benchmark's hardness is primarily driven by local landscape ruggedness and global aggregation structure, rather than by the connectivity patterns or regularization terms. Crucially, no single ablation restores the problem to the triviality of the fully simplified variant, and the combination of any two residual mechanisms is already sufficient to produce non-trivial landscapes at higher dimensions. This multi-mechanism character is precisely what the benchmark is designed to capture, and the ablation evidence confirms that the design achieves its intent.

The proposed benchmark addresses a recognized gap at the intersection of two research communities. Within continuous global optimization, the dominant benchmark suites including the CEC and BBOB families are composed primarily of synthetic analytical functions whose difficulty is controlled through transformation, rotation, and shifting, but which lack the semantic interpretability and state-dependent coupling of physically motivated problems. Within agricultural water management and irrigation scheduling, the available simulation-optimization frameworks are calibrated to specific field conditions and crop models and are therefore neither self-contained nor designed to be difficult for optimization algorithms per se. The present benchmark occupies a complementary position: it is analytically self-contained and fully deterministic, requires no external data, exhibits a rich and verifiable difficulty structure through the ablation protocol, and retains an agronomic interpretation that makes its components meaningful beyond pure algorithmic testing. The deliberate absence of a known closed-form global optimum further distinguishes it from classical synthetic benchmarks, making it a genuine black-box problem for which algorithmic progress can only be measured empirically.

4. Conclusions

This article has introduced a new continuous global optimization benchmark derived from the semantics of weather-coupled irrigation scheduling. The benchmark is formulated as a deterministic, box-constrained minimization problem in which the objective function integrates a recursive soil-water storage state, an oscillatory irrigation efficiency law, a multiplicative seasonal yield factor, neighbor-coupled hydraulic penalties, threshold-triggered

drainage losses, and a deceptive preferred-window regularization term. The problem is analytically self-contained: all exogenous weather and crop profiles are generated from the normalized stage coordinate, requiring no external datasets. Recommended problem dimensions span from $D = 24$ to $D = 100$, with the transition from moderate to high dimensionality producing a substantial and predictable increase in landscape difficulty.

The experimental study, conducted over 30 independent runs of eight established continuous optimizers across all recommended dimensions and eight problem variants, yields three principal conclusions. First, the full benchmark presents a genuinely multi-mechanism difficulty that no single state-of-the-art solver reliably resolves at $D \geq 70$: even the best-performing methods, mlshaderl and acor, achieve success rates below 25% at the highest dimension tested. Second, the ablation study identifies oscillatory efficiency modulation as the dominant source of landscape ruggedness, with multiplicative yield aggregation as the second most impactful mechanism; together, these two components account for the majority of the difficulty gap between the full and the simplified formulations. Third, the fully simplified variant, in which all embedded difficulty mechanisms are disabled, is solved to optimality by most methods across all dimensions, confirming that the observed difficulty in the full formulation is structural rather than numerical in origin. These results collectively validate the proposed benchmark as a meaningful and scalable tool for studying continuous optimization algorithms on non-separable, stateful, and rugged landscapes with a clear real-world interpretation.

Several directions of investigation emerge naturally from the present work and are left for future research. The current formulation employs a single synthetic weather profile generated from trigonometric functions of the normalized stage coordinate. A natural extension would introduce a parameterized family of profiles indexed, for instance, by the amplitude of the evapotranspiration cycle, the frequency of the rainfall component, or the magnitude of the tariff oscillation producing a suite of benchmark instances with controlled variability in the exogenous driving signals. Such a family would allow researchers to study how algorithm performance varies with the statistical properties of the weather input, a question of direct relevance to robust irrigation planning under forecast uncertainty.

A further generalization of considerable interest would introduce stochastic weather realizations, replacing the fixed analytical profiles with samples from a probability distribution calibrated to realistic seasonal variability. In this setting, the optimization problem becomes one of finding a decision vector that minimizes expected cost or maximizes robustness against worst-case scenarios, connecting the benchmark to the literature on stochastic and distributionally robust optimization. The recursive storage structure of the benchmark makes it particularly well suited for comparison with scenario-based and chance-constrained formulations. Along a related but distinct axis, the assumption that all irrigation decisions are made simultaneously at the beginning of the season could be relaxed. A dynamic extension, in which decisions are made sequentially with partial or noisy observations of the storage state between stages, would transform the benchmark into a sequential decision problem amenable to reinforcement learning, model predictive control, and Bayesian optimization methods, substantially broadening the community of researchers for whom the benchmark is relevant.

Because the benchmark is analytically defined and differentiable almost everywhere with the exception of the clipping and max operations it is also tractable for surrogate-assisted and gradient-based solvers, provided that subgradients or automatic differentiation are employed to handle the nonsmooth components. A systematic comparison of gradient-based methods, surrogate models, and derivative-free global optimizers on the same benchmark instances would provide empirical evidence on the trade-offs between first-order information exploitation and global exploration capacity in rugged, stateful landscapes. Complementing this direction, future work could apply established exploratory landscape analysis features such as fitness distance correlation, information content, and gradient homogeneity measures to the full and ablated formulations across all recommended dimensions. Such an analysis would provide a data-driven characterization of

how each difficulty mechanism shapes the search landscape and would connect the benchmark's properties to the broader landscape analysis literature, enabling more principled algorithm selection and configuration. Finally, the current formulation uses soft penalties to encode operational constraints such as the seasonal water budget and the peak hydraulic load. A future variant that introduces hard constraints on total water use or on per-stage application rates would transform the benchmark into a constrained continuous optimization problem, enabling comparison with constraint-handling techniques including penalty methods, feasibility-rule approaches, and augmented Lagrangian strategies, and further expanding the benchmark's utility as a testing ground for modern continuous optimization methodology.

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